

Q1 - 24 June - Shift 1

Let

$$S = \left\{ \theta \in [-\pi, \pi] - \left\{ \pm \frac{\pi}{2} \right\} : \sin \theta \tan \theta + \tan \theta = \sin 2\theta \right\}.$$

If $T = \sum_{\theta \in S} \cos 2\theta$, then $T + n(S)$ is equal

(A) $7 + \sqrt{3}$

(B) 9

(C) $8 + \sqrt{3}$

(D) 10

Space for your notes:

Q2 - 25 June - Shift 1

The number of values of x in the interval

$$\left(\frac{\pi}{4}, \frac{7\pi}{4} \right) \text{ for which } 14 \operatorname{cosec}^2 x - 2 \sin^2 x = 21$$

 $-4 \cos^2 x$ holds, is _____.

Space for your notes:

Q3 - 29 June - Shift 1

The number of elements in the set $S =$

$$\{ \theta \in [-4\pi, 4\pi] : 3 \cos^2 2\theta + 6 \cos 2\theta - 10 \cos^2 \theta + 5 = 0 \}$$

is _____.

Space for your notes:

Q4 - 29 June - Shift 1

The number of solutions of the equation

$$2\theta - \cos^2 \theta + \sqrt{2} = 0 \text{ is } R \text{ is equal to } \underline{\hspace{2cm}}.$$

Space for your notes:

Q5 - 29 June - Shift 2

The number of solutions of the equation $\sin x =$

$$\cos^2 x \text{ in the interval } (0, 10) \text{ is } \underline{\hspace{2cm}}.$$

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Answer Key

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Q1 (B)

Q2 (4)

Q3 (32)

Q4 (1)

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Q5 (4)

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Q1 (B)

$$\sin \theta \tan \theta + \tan \theta = \sin 2\theta$$

$$\tan \theta (\sin \theta + 1) = \frac{2 \tan \theta}{1 + \tan^2 \theta}$$

$$\tan \theta = 0 \Rightarrow \theta = -\pi, 0, \pi$$

$$(\sin \theta + 1) = 2 \cdot \cos^2 \theta = 2(1 + \sin \theta)(1 - \sin \theta)$$

$$\sin \theta = -1 \text{ which is not possible}$$

$$\sin \theta = \frac{1}{2} \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$n(s) = 5$$

$$T = \cos 0 + \cos 2\pi + \cos 2\pi + \cos \frac{\pi}{3} + \cos \frac{5\pi}{3}$$

$$T = 4$$

$$T + n(s) = 9$$

Q2 (4)

$$x \in \left(\frac{\pi}{4}, \frac{7\pi}{4} \right)$$

$$14 \operatorname{cosec}^2 x - 2 \sin^2 x = 21 - 4 \cos^2 x$$

$$= 21 - 4(1 - \sin^2 x)$$

$$= 17 + 4 \sin^2 x$$

$$14 \operatorname{cosec}^2 x - 6 \sin^2 x = 17$$

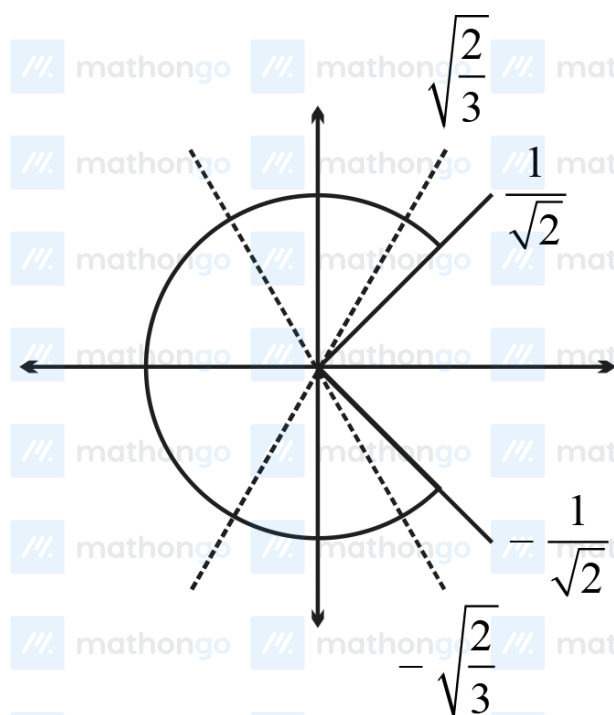
$$\text{let } \sin^2 x = p$$

$$\frac{14}{p} - 6p = 17 \Rightarrow 14 - 6p^2 = 17p$$

$$6p^2 + 17p - 14 = 0$$

$$p = -3.5, \frac{2}{3} \Rightarrow \sin^2 x = \frac{2}{3}$$

$$\Rightarrow \sin x = \pm \sqrt{\frac{2}{3}}$$



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Hints and Solutions

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$$3\cos^2 2\theta + 6 \cos 2\theta - 10 \cos^2 \theta + 5 = 0$$

$$3\cos^2 2\theta + 6 \cos 2\theta - 5(1 + \cos 2\theta) + 5 = 0$$

$$3\cos^2 2\theta + \cos 2\theta = 0$$

$$\cos 2\theta = 0 \text{ OR } \cos 2\theta = -1/3$$

$$\theta \in [-4\pi, 4\pi]$$

$$2\theta = (2n + 1) \cdot \frac{\pi}{2}$$

$$\therefore \theta = \pm \pi / 4, \pm 3\pi / 4, \dots, \pm 15\pi / 4$$

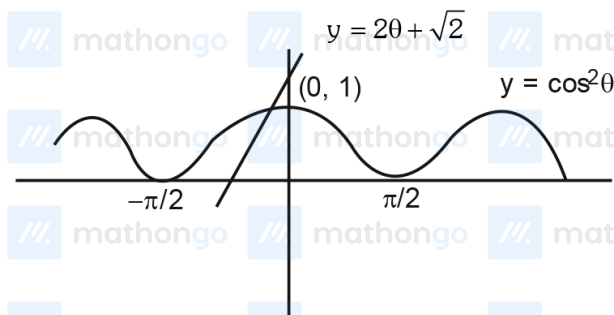
Similarly $\cos 2\theta = -1/3$ gives 16 solution

Q4 (1)

$$2\theta - \cos^2 \theta + \sqrt{2} = 0$$

$$\Rightarrow \cos^2 \theta = 2\theta + \sqrt{2}$$

$$y = 2\theta + \sqrt{2}$$



Both graphs intersect at one point.

Q5 (4)

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$$\sin^2 x + \sin x - 1 = 0$$

$$\sin x = \frac{-1 + \sqrt{5}}{2} = +ve$$

Only 4 roots

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